

Conversation 087:

Adiabatic Invariance Mathematical Treatment

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please provide me with a complete mathematical treatment of adiabatic invariance (to nth invariant)

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Adiabatic Invariance: Overview

Adiabatic invariance refers to quantities that remain constant when a system undergoes a slow, continuous change. This principle plays a key role in classical mechanics, quantum mechanics, and statistical physics.

For a physical system described by generalized coordinates q_i and momenta p_i , adiabatic invariants are quantities I_n such that:

$$\frac{dI_n}{dt} = 0 \quad (\text{approximately for slow variations}).$$

The simplest adiabatic invariant is the action integral:

$$I = \oint p dq,$$

which remains constant for a system with a slowly varying Hamiltonian. Higher-order invariants extend this principle.

Mathematical Framework

1. Hamiltonian Formalism

The system is described by a time-dependent Hamiltonian:

$$H(p, q, t),$$

where t is a slow time parameter. The equations of motion are:

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}.$$

2. Action-Angle Variables

For periodic systems, we introduce action-angle variables (J_i, θ_i) , where:

$$J_i = \frac{1}{2\pi} \oint p_i dq_i,$$

and θ_i are the conjugate angles. The Hamiltonian in these coordinates is:

$$H = H(J_1, J_2, \dots, t).$$

3. Adiabatic Invariance to the First Order

If the Hamiltonian changes slowly, $\frac{\partial H}{\partial t} \ll 1$, the action variables J_i are adiabatically invariant:

$$\frac{dJ_i}{dt} = 0.$$

Proof:

1. Rewrite the Hamiltonian equations of motion in terms of J_i and θ_i .
2. Show that the time dependence of H does not affect the periodicity of the system, leading to the conservation of J_i .

4. Adiabatic Invariance to Higher Orders (nth Invariant)

Higher-order adiabatic invariants account for non-periodic or weakly perturbed systems. These are derived from the expansion of the Hamiltonian in terms of a small parameter ϵ , representing the slow variation.

$$H = H_0 + \epsilon H_1 + \epsilon^2 H_2 + \dots$$

1. Expand the equations of motion:

Expand $q_i(t)$ and $p_i(t)$ in powers of ϵ :

$$q_i = q_i^{(0)} + \epsilon q_i^{(1)} + \epsilon^2 q_i^{(2)} + \dots$$

$$p_i = p_i^{(0)} + \epsilon p_i^{(1)} + \epsilon^2 p_i^{(2)} + \dots$$

2. Apply perturbation theory:

Substitute these expansions into the Hamiltonian equations and solve iteratively for each order ϵ^n .

3. Nth adiabatic invariant:

The n -th adiabatic invariant I_n is obtained by isolating terms proportional to ϵ^n and integrating over one period of the unperturbed motion:

$$I_n = \oint F_n(p, q) dq,$$

where $F_n(p, q)$ is a term derived from the n -th order perturbative correction.

5. Quantum Mechanical Treatment

In quantum systems, the adiabatic theorem states that if a Hamiltonian changes slowly compared to the system's natural frequencies, the quantum numbers associated with the system remain invariant.

The quantum adiabatic invariant is linked to the Bohr-Sommerfeld quantization condition:

$$\oint p \, dq = nh,$$

where n is a quantum number, h is Planck's constant, and the integral remains invariant under slow changes.

Practical Applications

1. Classical Mechanics:

- Pendulum with slowly varying length: $I = \oint p \, dq$ remains constant as the pendulum's length changes.

2. Statistical Mechanics:

- Slowly compressing a gas: The product PV^γ (where γ is the adiabatic index) is an invariant.

3. Plasma Physics:

- Magnetic flux conservation: $\Phi = \oint \mathbf{B} \cdot d\mathbf{A}$ is an adiabatic invariant for charged particles in magnetic fields.

4. Quantum Mechanics:

- Population of quantum states: Energy levels $E_n(t)$ adjust slowly, preserving quantum numbers.

If you would like a more detailed explanation of any particular derivation or application, feel free to ask!

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Nth adiabatic invariant: The

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To provide a detailed explanation of how the n -th adiabatic invariant I_n is constructed, let's delve into the mathematical framework step by step.

General Idea of Higher-Order Adiabatic Invariance

For a system with a slowly varying Hamiltonian $H(p, q, t)$, higher-order adiabatic invariants are corrections that remain approximately conserved even when deviations from exact periodicity or linearity are considered. These corrections arise from perturbation theory, where the Hamiltonian is expanded in powers of a small parameter ϵ , representing the slowness of the variation.

1. Expansion of the Hamiltonian

The Hamiltonian is written as:

$$H(p, q, t) = H_0(p, q) + \epsilon H_1(p, q, t) + \epsilon^2 H_2(p, q, t) + \dots,$$

where:

- $H_0(p, q)$ represents the unperturbed Hamiltonian (time-independent and periodic).
- $H_1, H_2, \dots$ are perturbative corrections proportional to $\epsilon, \epsilon^2, \dots$.

2. Perturbative Expansion of Motion

The generalized coordinates $q(t)$ and momenta $p(t)$ are also expanded in powers of ϵ :

$$\begin{aligned} q(t) &= q^{(0)}(t) + \epsilon q^{(1)}(t) + \epsilon^2 q^{(2)}(t) + \dots, \\ p(t) &= p^{(0)}(t) + \epsilon p^{(1)}(t) + \epsilon^2 p^{(2)}(t) + \dots. \end{aligned}$$

Here:

- $q^{(0)}, p^{(0)}$: Motion under H_0 (periodic).
- $q^{(1)}, p^{(1)}$, etc.: Corrections due to the perturbations.

3. Equation of Motion for Perturbations

The equations of motion arise from Hamilton's equations:

$$\dot{q} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial q}.$$

Substituting the expansions of q, p , and H , we get a hierarchy of equations for different orders of ϵ :

- **Zeroth-order (unperturbed motion):**

$$\dot{q}^{(0)} = \frac{\partial H_0}{\partial p^{(0)}}, \quad \dot{p}^{(0)} = -\frac{\partial H_0}{\partial q^{(0)}}.$$

This describes periodic motion.

- **First-order corrections:**

$$\dot{q}^{(1)} = \frac{\partial H_1}{\partial p^{(0)}} + \frac{\partial H_0}{\partial p^{(1)}}, \quad \dot{p}^{(1)} = -\frac{\partial H_1}{\partial q^{(0)}} - \frac{\partial H_0}{\partial q^{(1)}}.$$

- **Second-order corrections:**

Include terms involving H_2 and nonlinear combinations of $q^{(1)}, p^{(1)}$.

4. Action Integral and Adiabatic Invariants

The key idea of adiabatic invariance is that certain combinations of $q(t)$ and $p(t)$ remain approximately constant under slow variations. At the n -th order, the invariant I_n is defined by:

$$I_n = \oint F_n(p, q) dq,$$

where:

- $\oint$ denotes integration over one full cycle of the unperturbed motion.
- $F_n(p, q)$ is a function derived from the n -th order perturbative correction to $H(p, q, t)$.

5. Derivation of $F_n(p, q)$

1. **Perturbative correction:** From the n -th order equations of motion, the deviation from the periodic trajectory $(q^{(0)}, p^{(0)})$ is computed as $(q^{(n)}, p^{(n)})$. These deviations encode the effect of higher-order terms.
2. **Periodic averaging:** Over one period of the unperturbed motion $T = 2\pi/\omega$, the perturbative contributions $q^{(n)}(t)$ and $p^{(n)}(t)$ vary slowly. The integral averages out the fast oscillations.
3. **Construction of $F_n(p, q)$:**
The perturbation term $F_n(p, q)$ is extracted from the Hamiltonian corrections and incorporates:
 - Time-dependent terms $\frac{\partial H_n}{\partial t}$.
 - Couplings between the zeroth-order terms $q^{(0)}, p^{(0)}$ and the n -th order corrections.
4. **Action integral:** $F_n(p, q)$ is then integrated over one full period of $q^{(0)}$, resulting in the n -th adiabatic invariant.

6. Practical Example

For a simple harmonic oscillator with slowly varying frequency $\omega(t)$, the zeroth-order Hamiltonian is:

$$H_0 = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2.$$

The zeroth-order action is:

$$I_0 = \oint p dq = \frac{E}{\omega},$$

where E is the total energy.

If the frequency $\omega(t)$ changes slowly (adiabatic variation), the higher-order corrections $F_n(p, q)$ include terms like $\frac{d\omega}{dt}$ and its derivatives, leading to higher-order invariants.

Conclusion

The n -th adiabatic invariant I_n is obtained by systematically isolating the n -th order perturbative terms and averaging them over one full cycle of unperturbed motion. The resulting integral encodes the slow variation

of the system and provides a deeper understanding of how invariants emerge under adiabatic changes.

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provide more detail on coupling terms and how they might shape F_n

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Coupling Terms and Their Role in $F_n(p, q)$

Coupling terms arise naturally in the perturbative expansion of the Hamiltonian and equations of motion. These terms represent the interaction between different degrees of freedom or corrections at various orders, affecting the n -th order perturbative contribution $F_n(p, q)$. Here's a detailed analysis of how coupling terms influence $F_n(p, q)$.

1. Hamiltonian Expansion

The total Hamiltonian is expressed as:

$$H(p, q, t) = H_0(p, q) + \epsilon H_1(p, q, t) + \epsilon^2 H_2(p, q, t) + \dots,$$

where:

- $H_0(p, q)$: Unperturbed (periodic) Hamiltonian.
- $H_1(p, q, t)$: First-order perturbation, typically time-dependent.
- $H_2(p, q, t)$: Second-order correction, includes nonlinear effects and coupling between H_0 and H_1 .

Each perturbative term H_n generates coupling terms that influence the equations of motion and ultimately shape $F_n(p, q)$.

2. Coupling in the Equations of Motion

The equations of motion are derived from Hamilton's equations:

$$\dot{q} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial q}.$$

Substituting the expansion of H , we get a hierarchy of equations:

- **Zeroth order** ($\mathcal{O}(1)$):

$$\dot{q}^{(0)} = \frac{\partial H_0}{\partial p^{(0)}}, \quad \dot{p}^{(0)} = -\frac{\partial H_0}{\partial q^{(0)}}.$$

- **First order** ($\mathcal{O}(\epsilon)$):

$$\dot{q}^{(1)} = \frac{\partial H_1}{\partial p^{(0)}} + \frac{\partial H_0}{\partial p^{(1)}}, \quad \dot{p}^{(1)} = -\frac{\partial H_1}{\partial q^{(0)}} - \frac{\partial H_0}{\partial q^{(1)}}.$$

Here, H_1 introduces coupling between the zeroth-order dynamics $(q^{(0)}, p^{(0)})$ and the first-order corrections $(q^{(1)}, p^{(1)})$.

- **Second order** ($\mathcal{O}(\epsilon^2)$):

$$\begin{aligned} \dot{q}^{(2)} &= \frac{\partial H_2}{\partial p^{(0)}} + \frac{\partial H_1}{\partial p^{(1)}} + \frac{\partial H_0}{\partial p^{(2)}}, \\ \dot{p}^{(2)} &= -\frac{\partial H_2}{\partial q^{(0)}} - \frac{\partial H_1}{\partial q^{(1)}} - \frac{\partial H_0}{\partial q^{(2)}}. \end{aligned}$$

These equations include coupling terms involving:

- First-order corrections $(H_1, q^{(1)}, p^{(1)})$.
- Second-order corrections $(H_2, q^{(2)}, p^{(2)})$.
- Mixed terms $(H_0 \times H_1, H_1 \times H_2)$.

3. Shape of $F_n(p, q)$

The function $F_n(p, q)$, derived from the n -th order corrections, is shaped by the coupling terms at that order. Specifically:

(a) Contribution from H_n :

The n -th order perturbation $H_n(p, q, t)$ contributes directly to $F_n(p, q)$ through terms like:

$$F_n(p, q) \sim \frac{\partial H_n}{\partial q^{(0)}} \quad \text{and} \quad \frac{\partial H_n}{\partial p^{(0)}}.$$

For example, if H_n includes time-dependent terms such as $H_n \sim f(q, p) \cos(\omega t)$, then F_n will have periodic components influenced by $f(q, p)$.

(b) Coupling Between Orders:

Coupling terms from lower orders (e.g., $H_0 \times H_1$, $H_1 \times H_2$) introduce nonlinear dependencies. For instance:

- First-order corrections ($q^{(1)}, p^{(1)}$) are driven by the zeroth-order dynamics H_0 and the perturbation H_1 .
- At the second order, H_2 couples H_0 , H_1 , and the first-order corrections.

Thus, $F_n(p, q)$ includes contributions from terms like:

$$F_n(p, q) \sim \int \left(\frac{\partial H_0}{\partial p} \frac{\partial H_1}{\partial q} - \frac{\partial H_0}{\partial q} \frac{\partial H_1}{\partial p} \right) dt.$$

(c) Time Dependence and Resonances:

If $H_n(p, q, t)$ has explicit time dependence (e.g., $H_n \sim A \sin(\omega t)$), resonances may appear. These resonances modulate $F_n(p, q)$ through integrals of the form:

$$\int f(p, q) \cos(\omega t) dt \quad \text{or} \quad \int g(p, q) \sin(\omega t) dt,$$

leading to frequency-dependent corrections.

4. Final Expression for $F_n(p, q)$

After collecting all contributions, $F_n(p, q)$ takes the form:

$$F_n(p, q) = \sum_{m=0}^n [C_m(p, q) \cos(m\omega t) + D_m(p, q) \sin(m\omega t)],$$

where:

- $C_m(p, q)$, $D_m(p, q)$: Coefficients determined by coupling terms and perturbations.
- $m\omega$: Harmonics arising from the periodicity of $q^{(0)}$ and $p^{(0)}$.

The action integral is then:

$$I_n = \oint F_n(p, q) dq,$$

where the integral over one period averages out fast oscillations and isolates the slowly varying components.

5. Example: Harmonic Oscillator with Slowly Varying Frequency

Consider a harmonic oscillator with a time-dependent frequency $\omega(t) = \omega_0 + \epsilon\omega_1(t)$. The unperturbed Hamiltonian is:

$$H_0 = \frac{p^2}{2m} + \frac{1}{2}m\omega_0^2 q^2.$$

The first-order perturbation is:

$$H_1 = \frac{1}{2} m \left(\omega_1^2(t) - \omega_0^2 \right) q^2.$$

Coupling terms introduce corrections like:

$$F_1(p, q) \sim m q^2 \frac{d\omega_1}{dt},$$

which affect the first-order invariant I_1 . At higher orders, nonlinear couplings (e.g., q^4 terms) appear, modifying $F_n(p, q)$.

If you have specific systems or coupling mechanisms in mind, I can tailor the treatment further!

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